

Sieve Plate Distillation

Name: Purva Kaushalbhai Shah

Group members name:

Rachna Durgam

Om Gupta

Table of Contents:

Objective

Theory and Background

Known parameters

Precautions

Formulae

Readings

Results and Discussion

Conclusion

References

Objective:

The aim of this experiment is to determine the theoretical number of plates for a sieve plate distillation column and find the overall efficiency.

Theory and Background:

Distillation is a common method used to separate liquids based on their volatility. The component that evaporates more easily (higher volatility) turns into vapour first, while the less volatile one stays in the liquid phase. The bigger the difference in volatility, the easier it is to separate them. If the difference is small, simple distillation doesn't work well, so we use multistage distillation. In this process, sieve plates inside the column help by allowing vapour and liquid to mix properly at each stage, improving separation. The design of these plates, their hole size, and the flow of liquid and vapour all affect how well the separation works.

One way to get maximum separation is through total reflux, where all the condensed vapour is returned to the column, and no product is removed. Under these conditions, the operating lines of the distillation column align with the 45° diagonal in an equilibrium diagram, making it easier to calculate the number of stages needed using the McCabe-Thiele method.

Reflux Ratio

The reflux ratio (R) is the amount of liquid sent back into the column compared to the liquid collected as the final product. A higher reflux ratio improves separation by increasing the number of vapour-liquid contact stages, but it also uses more energy. In total reflux conditions ($R = \infty$), all condensed vapours are returned to the column, ensuring maximum separation efficiency. In real operations, a balance is needed—too low a reflux ratio leads to poor separation, while too high a ratio wastes energy. The goal is to find the best reflux ratio that achieves good separation without excessive energy costs.

Efficiency

The efficiency of the column is measured by comparing the actual number of plates to the theoretical number required.

$$\text{Efficiency} = \frac{\text{Number of theoretical trays}}{\text{Number of practical trays}} \times 100$$

The efficiency tells us how efficient our distillation column is as it accounts for practical inefficiencies such as liquid distribution, imperfect vapour-liquid contact, etc.

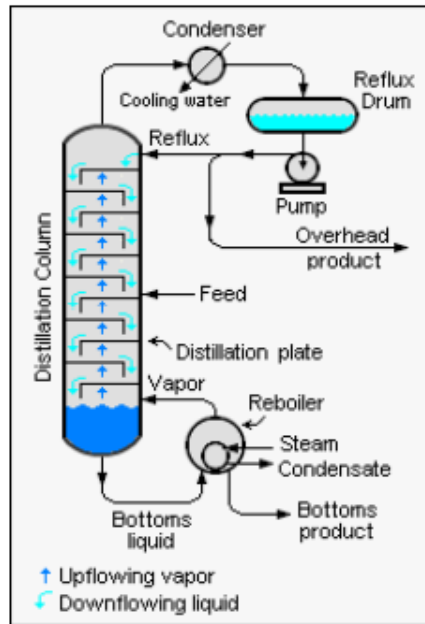


Figure 0: Schematics of the experiment [4]



Figure 1: Experimental set-up

Apparatus:

1. Sieve plate distillation column
2. Digital refractometer
3. Measuring cylinders
4. Level indicator

5. N-propanol
6. Iso-propanol

Procedure:

1. Equimolar solution of iso-propanol (155 mL) and n-propanol (150 mL) is prepared.
2. Check the RI of the solution using a refractometer.
3. Turn on the heater.
4. Set the temperature of heater to 135°C.
5. Allow the system to reach equilibrium, in this process vapours coming out from the solution will get condensed and come out as a distillate.
6. Once distillation begins, record the refractive index (RI) of the distillate at $t=0$ minutes.
7. Simultaneously, measure and record the RI of the residue in the cylinder.
8. After every 5 minutes, note down the RI values of condensate and residue.
9. Continue the process until you get steady RI values.
10. Calculate the mole fraction fraction value in each phase using the calibration curve of RI vs X.
11. From the given standard X-Y data, plot the equilibrium curve.
12. . Draw the maximum possible perpendicular triangles between the XD and XW values.
13. The total number of triangles will give the theoretical number of plates

Known parameters:

- *Boiling point of iso – propanol = 82°C*
- *Boiling point of n – propanol = 97°C*
- *RI of n – propanol: 1.3733*
- *RI of iso – propanol: 1.3812*

Precautions:

- Wear gloves while collecting the samples of distillate and residue, as the temperature of both the components is high.
- Collect a small amount of sample (few drops) to prevent the change in the composition of the binary mixture.
- While taking reading using a refractometer cover the sample by hand otherwise surroundings will affect the refractive index.
- Clean the refractometer lens after every reading.
- Wash the beakers used for collecting samples after every reading to get exact readings for small changes in RI.

Formulae:

- *RI to mole fraction: For 2 species A and B, if A is more volatile*

- $X_A = \frac{\text{reading obtained} - \text{RI of pure A}}{|\text{RI of pure A} - \text{RI of pure B}|}$
- For equilibrium curve we use Roul't's law:
 - $y_i P = x_i P_i^{\text{sat}}$
- For P_i^{sat} we used Antoine equation
 - $\ln P_i^{\text{sat}} = A - \frac{B}{T+C}$, where A, B, C are Antoine constants.
 - T is the temperature in K
- Actual number of plates in the column : 4

Readings:

TABLE 1: TABLE CONTAINING EXPERIMENTAL DATA

Sr. no	Time (min)	Top Product Refractive Index (Rd)	Bottom Product Refractive Index (Rr)
1	0	1.3768	1.3799
2	5	1.3763	1.3801
3	10	1.3768	1.38
4	15	1.3778	1.38
5	20	1.377	1.3802
6	25	1.3772	1.3801
7	30	1.377	1.3804
8	35	1.3776	1.3807
9	40	1.3777	1.3809
10	45	1.3778	1.3813
11	50	1.3779	1.3813
12	55	1.3779	1.3814
13	60	1.3781	1.3815
14	65	1.3763	1.3804
15	70	1.3763	1.3804

TABLE 2: TABLE CONTAINING CALCULATED DATA OF MOLE FRACTION AND REFRACTIVE INDEX

Sr.	Time	Top Product	Bottom Product	Mole Fraction of	Mole Fraction of
-----	------	-------------	----------------	------------------	------------------

no	(min)	Refractive Index (Rd)	Refractive Index (Rr)	iso-propanol in Distillate (Xd)	iso-propanol in Residue (Xr)
1	0	1.3768	1.3799	0.5569620253	0.164556962
2	5	1.3763	1.3801	0.6202531646	0.1392405063
3	10	1.3768	1.38	0.5569620253	0.1518987342
4	15	1.3778	1.38	0.4303797468	0.1518987342
5	20	1.377	1.3802	0.5316455696	0.1265822785
6	25	1.3772	1.3801	0.5063291139	0.1392405063
7	30	1.377	1.3804	0.5316455696	0.1012658228
8	35	1.3776	1.3807	0.4556962025	0.06329113924
9	40	1.3777	1.3809	0.4430379747	0.03797468354
10	45	1.3778	1.3813	0.4303797468	0.01265822785
11	50	1.3779	1.3813	0.417721519	0.01265822785
12	55	1.3779	1.3814	0.417721519	0.0253164557
13	60	1.3781	1.3815	0.3924050633	0.03797468354
14	65	1.3763	1.3804	0.6202531646	0.1012658228
15	70	1.3763	1.3804	0.6202531646	0.1012658228

TABLE 3: TABLE CONTAINING EQUILIBRIUM CURVE DATA AND Y=X LINE DATA

X	Y_{eq}	Y
0	0	0
0.0575	0.11	0.0575
0.061	0.111	0.061
0.1455	0.2325	0.1455
0.2285	0.351	0.2285
0.3095	0.4435	0.3095
0.3125	0.45	0.3125
0.42	0.5545	0.42
0.4355	0.5725	0.4355
0.519	0.66	0.519
0.631	0.748	0.631

0.735	0.8225	0.735
0.7675	0.8495	0.7675
0.8585	0.9175	0.8585
0.91	0.9525	0.91
1	1	1

Antoine equation constants:

$$A = 4.861$$

$$B = 1357.427$$

$$C = -75.814$$

$$T = 273 + 100 \text{ K}$$

$$P^{sat} = 1.34 \text{ bar}$$

Results and discussion:

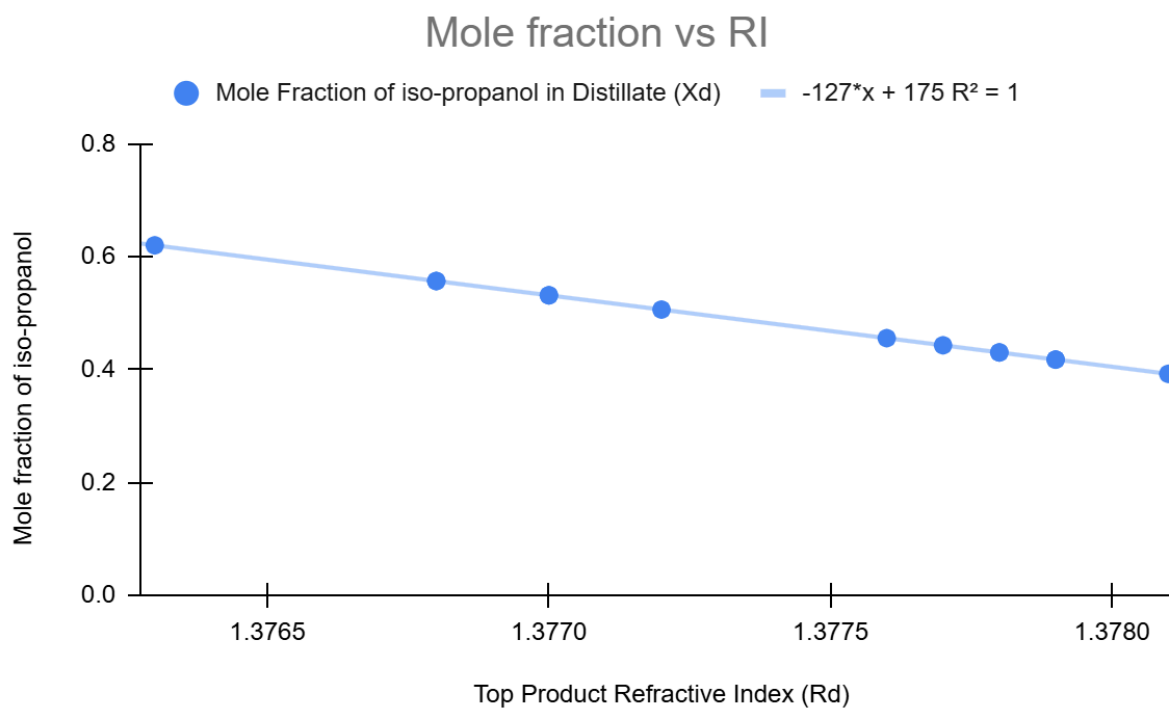


Figure 2: Figure showing the plot between Mole fraction of iso-propanol in distillate vs RI of the distillate

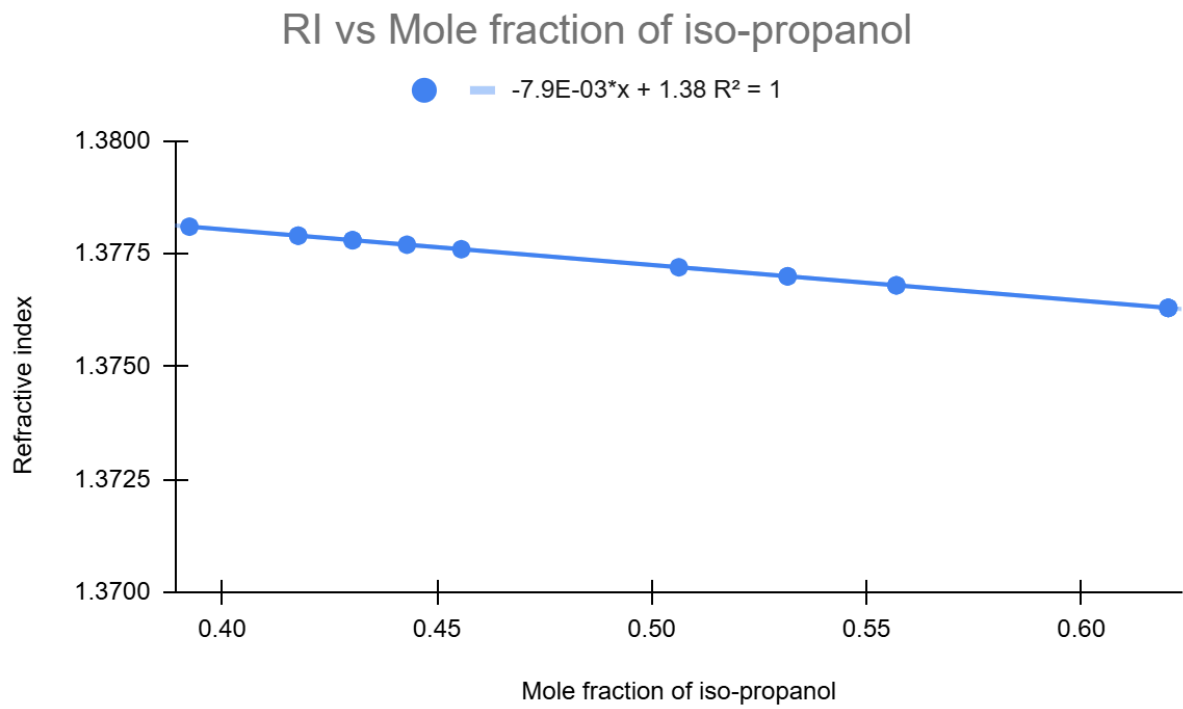


Figure 3: Figure showing the plot between RI of the distillate vs Mole fraction of iso-propanol in distillate

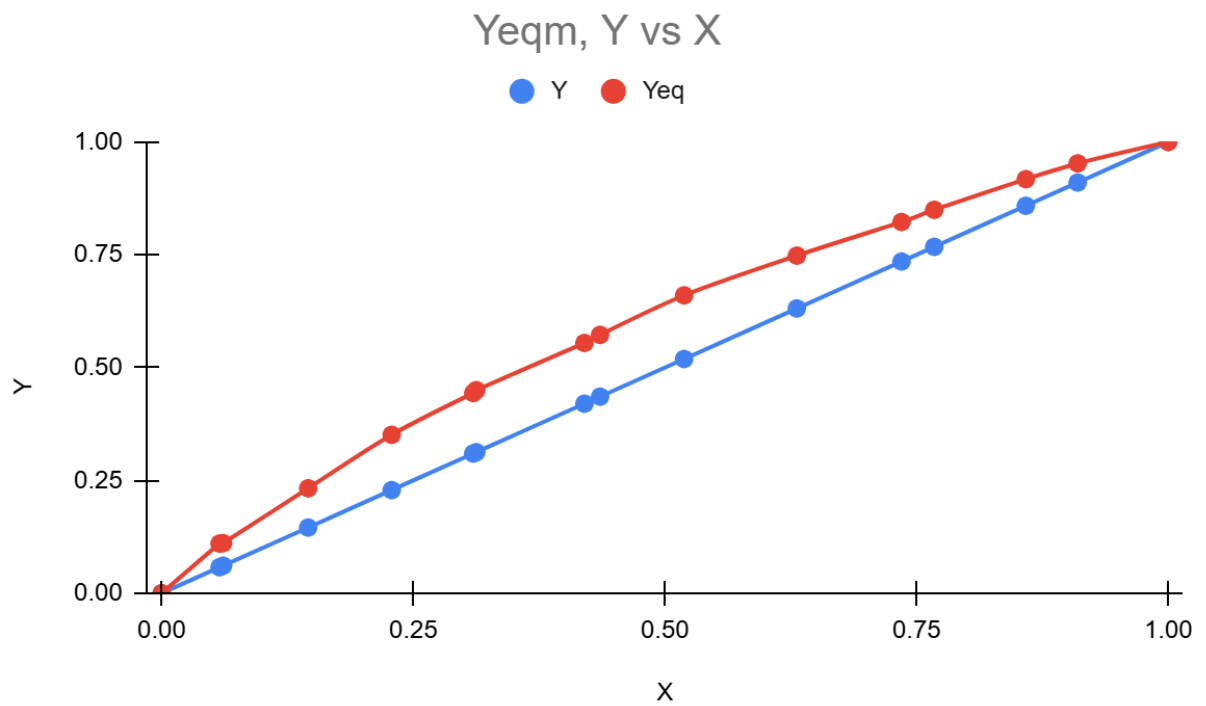


Figure 4: Figure showing the plot between Y_{eq} , Y vs X

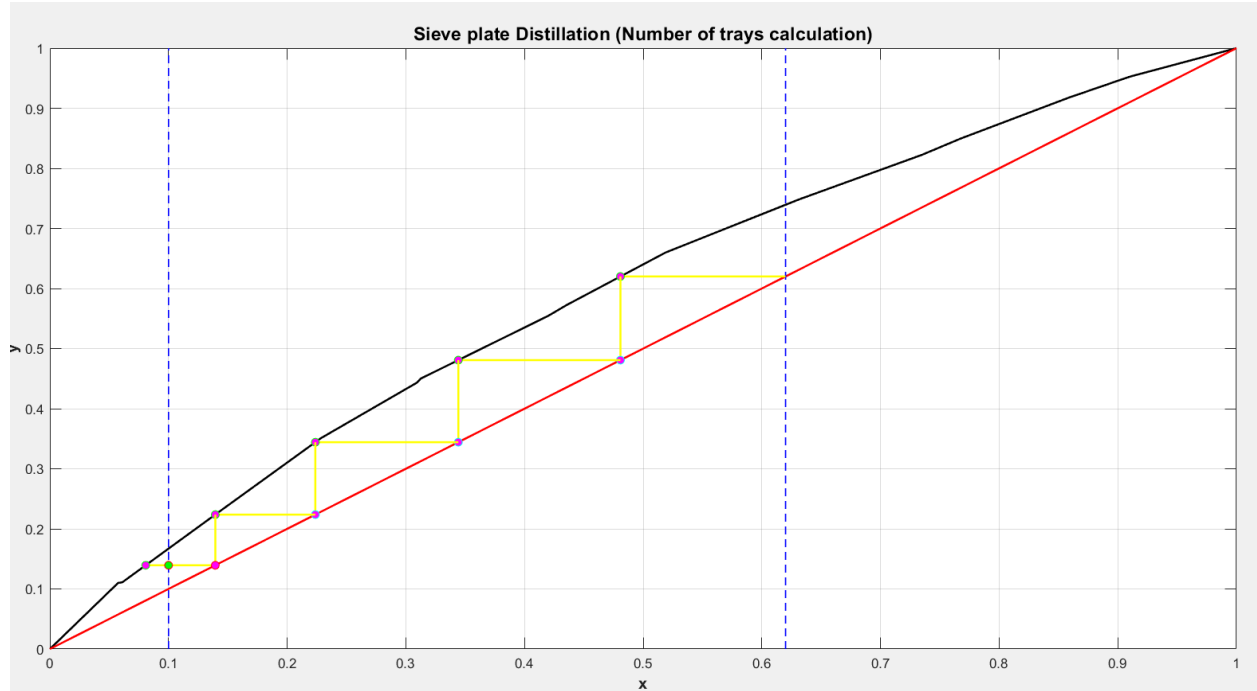


Figure 5: Figure showing the number of trays calculated from McCabe-Thiele method

Equilibrium mole fraction of iso-propanol in distillate = 0.62

Equilibrium mole fraction of iso-propanol in residue = 0.10

N_p = number of theoretical plates

From the plot $N_p + 1 = 4.67 \Rightarrow N_p = 3.67$

Number of theoretical sieve plates = 3.67

Number of actual sieve plates = 4

$$\text{Overall efficiency} = \frac{\text{Number of theoretical sieve plates}}{\text{Number of actual sieve plates}} \times 100 = \frac{3.67}{4} \times 100 = 91.75\%$$

In figure 3, we can see that as the mole fraction decreases and the refractive index increases. The equation of mole fraction vs RI is calibrated as : $y = -0.0079x + 1.38$, where y is the refractive index and x is the mole fraction of iso-propanol in the distillate.

In figure 5, The black line is our Equilibrium curve, the Red line is the Y=X curve, and yellow lines are the tray calculations. Blue dotted lines tell us where to start from and where to end. From matlab code, we get the number of trays to be 4.67.

Conclusion:

This experiment successfully determined the theoretical number of sieve plates needed to separate an equimolar n-propanol and iso-propanol mixture using a sieve plate distillation

column under total reflux conditions. Based on refractive index measurements and McCabe-Thiele analysis, the theoretical number of plates was found to be 3.67, while the actual number of plates in the column was 4. This resulted in an overall column efficiency of 91.75%, indicating efficient vapour-liquid contact and mass transfer.

The slight difference between the theoretical and actual number of plates is due to practical inefficiencies such as minor vapour losses, liquid flow irregularities, and slight deviations in equilibrium. However, the high efficiency of 91.75% suggests that the sieve plate design provided effective separation. The equilibrium mole fractions—0.62 in the distillate and 0.10 in the residue—further confirm the success of the process. Additionally, the calibration equation $y = -0.0079x + 1.38$ from Figure 3 shows a clear relationship between mole fraction and refractive index, reinforcing the observed separation trend.

Optimizing distillation efficiency is essential for reducing energy consumption, improving product purity, and lowering operational costs in industrial applications. Future studies could explore different feed compositions or operating conditions to enhance separation performance. These insights are valuable for designing more efficient and sustainable distillation systems.

Sources of errors:

- The distillation set up may not be cleaned properly before starting the experiment.
- There can be temperature fluctuations which can affect equilibrium.
- The refractometer may not have been cleaned properly.
- The beaker in which we are collecting small samples may not have been cleaned properly.

References:

- [1] "Isopropyl Alcohol." *Webbook.nist.gov*, webbook.nist.gov/cgi/cbook.cgi?ID=C67630&Mask=4&Type=ANTOINE&Plot=on#ANTOINE.
- [2] 'Lab Manual' provided in the classroom.
- [3] "What Is Sieve Tray Distillation Column?" *Sutongtechnology.com*, 2024, www.sutongtechnology.com/blog/what-is-sieve-tray-distillation-column.html.
- [4] "Distillation", <https://en.wikipedia.org/wiki/Distillation>
- [5] Mass Transfer Operations by Robert E. Treybal

MATLAB CODE, EXCEL SHEET is submitted in the submission separately.